

Theorem 11.3.5. (Erdős & Rényi 1963)

With probability 1, a random graph $G \in \mathcal{G}(\aleph_0, p)$ with $0 < p < 1$ is isomorphic to the Rado graph R .

Proof. Given fixed disjoint finite sets $U, W \subseteq \mathbb{N}$, the probability that a vertex $v \notin U \cup W$ is not joined to $U \cup W$ as expressed in property (*) of Chapter 8.3 (i.e., is not joined to all of U or is joined to some vertex in W) is some number $r < 1$ depending only on U and W . The probability that none of k given vertices v is joined to $U \cup W$ as in (*) is r^k , which tends to 0 as $k \rightarrow \infty$. Hence the probability that all the (infinitely many) vertices outside $U \cup W$ fail to witness (*) for these sets U and W is 0. (8.3.1)

Now there are only countably many choices for U and W as above. Since the union of countably many sets of measure 0 again has measure 0, the probability that (*) fails for any sets U and W is still 0. Therefore G satisfies (*) with probability 1. By Theorem 8.3.1 this means that, almost surely, $G \cong R$. $\square$

How can we make sense of the paradox that the result of infinitely many independent choices can be so predictable? The answer, of course, lies in the fact that the uniqueness of R holds only up to isomorphism. Now, constructing an automorphism for an infinite graph with property (*) is a much easier task than finding one for a finite random graph, so in this sense the uniqueness is no longer that surprising. Viewed in this way, Theorem 11.3.5 expresses not a lack of variety in infinite random graphs but rather the abundance of symmetry that glosses over this variety when the graphs $G \in \mathcal{G}(\aleph_0, p)$ are viewed only up to isomorphism.

11.4 Threshold functions and second moments

The results we saw in Section 11.3 have an interesting common feature: the values of p played no role as long as they were constant, that is, independent of n . For example, if almost every graph in $\mathcal{G}(n, p)$ with $p = 0.99$ had the property considered, then the same was true for $p = 0.01$. How could this happen?

Such insensitivity of our random model to changes of p was certainly not intended. For most properties, however, the critical order of magnitude of p around which the property will ‘just’ occur or not occur simply lies below any constant value of p : it is a function of n tending to zero as $n \rightarrow \infty$. In the proof of Erdős’s theorem, for example, this critical probability for the two properties we were trying to relate was $p(n) = 1/n$.

threshold
function

Let us call a positive real function $t = t(n)$ a *threshold function* for a graph property $\mathcal{P}$ if the following holds for all $p = p(n)$ and $G \in \mathcal{G}(n, p)$:

$$\lim_{n \rightarrow \infty} \mathbb{P}[G \in \mathcal{P}] = \begin{cases} 0 & \text{if } p/t \rightarrow 0 \text{ as } n \rightarrow \infty \\ 1 & \text{if } p/t \rightarrow \infty \text{ as } n \rightarrow \infty. \end{cases}$$

If $\mathcal{P}$ has a threshold function t , then clearly any positive multiple ct of t is also a threshold function for $\mathcal{P}$; thus, threshold functions in the above sense are only ever unique up to a multiplicative constant.⁶

Bollobás & Thomason (1987) have shown that, trivial exceptions aside, all *increasing* graph properties have threshold functions, properties that are closed under the addition of edges. Properties of the form $\{G \mid G \supseteq H\}$, with H fixed, are a common example; we shall compute their threshold functions in this section.

For the purpose of computing its threshold function, it is convenient to cast the graph property $\mathcal{P}$ considered in the form

$$\mathcal{P} = \{G \mid X(G) \geq 1\},$$

$X \geq 0$

where $X \geq 0$ is a suitable random variable on $\mathcal{G}(n, p)$. For example, we could take the indicator random variable of $\mathcal{P}$ on $\mathcal{G}(n, p)$. But other choices of X are allowed too; if $\mathcal{P}$ is connectedness, for example, we might have $X(G)$ count the number of spanning trees of G .

How could we prove that some given t is a threshold function of $\mathcal{P}$? Any such proof will consist of two parts: a proof that almost no $G \in \mathcal{G}(n, p)$ has $\mathcal{P}$ when p is small compared with t , and one showing that almost every G has $\mathcal{P}$ when p is large.

Since $X \geq 0$, we may use Markov's inequality for the first part of the proof and find an upper bound for $\mathbb{E}(X)$ instead of $\mathbb{P}[X \geq 1]$: if $\mathbb{E}(X)$ is much smaller than 1 then $X(G)$ can be at least 1 only for few $G \in \mathcal{G}(n, p)$, and for almost no G if $\mathbb{E}(X) \rightarrow 0$ as $n \rightarrow \infty$. Besides, the expectation is much easier to calculate than probabilities: without worrying about such things as independence or incompatibility of events, we may compute the expectation of a sum of random variables – for example, of indicator random variables – simply by adding up their individual expected values.

For the second part of the proof, things are more complicated. In order to show that $\mathbb{P}[X \geq 1]$ is large, it is not enough to bound $\mathbb{E}(X)$ from below: since X is not bounded above, $\mathbb{E}(X)$ may be large simply because X is very large on just a few graphs G – so X may still be zero

⁶ Our notion of threshold reflects only the crudest interesting level of screening: for some properties, such as connectedness, one can define sharper thresholds where the constant factor is crucial.

for most $G \in \mathcal{G}(n, p)$.⁷ In order to prove that $\mathbb{P}[X \geq 1] \rightarrow 1$, we thus have to show that this cannot happen, i.e., that X does not deviate a lot from its mean too often.

The following tool from probability theory achieves just that. As is customary, we write

$$\mu := \mathbb{E}(X) \quad \mu$$

and define $\sigma \geq 0$ by setting

$$\sigma^2 := \mathbb{E}((X - \mu)^2). \quad \sigma^2$$

This quantity σ^2 is called the *variance* or *second moment* of X ; by definition, it is a measure of how much X deviates from its mean. Since $\mathbb{E}$ is linear, the defining term for σ^2 expands to

$$\sigma^2 = \mathbb{E}(X^2 - 2\mu X + \mu^2) = \mathbb{E}(X^2) - \mu^2.$$

Note that μ and σ^2 always refer to a random variable on some fixed probability space. In our setting, where we consider the spaces $\mathcal{G}(n, p)$, both quantities are functions of n .

The following lemma says exactly what we need: that X cannot deviate a lot from its mean too often.

Lemma 11.4.1. (Chebyshev's Inequality)

For all real $\lambda > 0$,

$$\mathbb{P}[|X - \mu| \geq \lambda] \leq \sigma^2/\lambda^2.$$

Proof. By Lemma 11.1.5 and definition of σ^2 , (11.1.5)

$$\mathbb{P}[|X - \mu| \geq \lambda] = \mathbb{P}[(X - \mu)^2 \geq \lambda^2] \leq \sigma^2/\lambda^2.$$

□

For a proof that $X(G) \geq 1$ for almost all $G \in \mathcal{G}(n, p)$, Chebyshev's inequality can be used as follows:

Lemma 11.4.2. If $\mu > 0$ for all large enough n , and $\sigma^2/\mu^2 \rightarrow 0$ as $n \rightarrow \infty$, then $X(G) > 0$ for almost all $G \in \mathcal{G}(n, p)$.

⁷ For some p between n^{-1} and $(\log n)n^{-1}$, for example, almost every $G \in \mathcal{G}(n, p)$ has an isolated vertex (and hence no spanning tree), but its expected number of spanning trees tends to infinity with n . See the Exercise 11 for details.

Proof. Any graph G with $X(G) = 0$ satisfies $|X(G) - \mu| = \mu$. Hence Lemma 11.4.1 implies with $\lambda := \mu$ that

$$\mathbb{P}[X = 0] \leq \mathbb{P}[|X - \mu| \geq \mu] \leq \sigma^2/\mu^2 \xrightarrow{n \rightarrow \infty} 0.$$

Since $X \geq 0$, this means that $X > 0$ almost surely, i.e. that $X(G) > 0$ for almost all $G \in \mathcal{G}(n, p)$. $\square$

As the main result of this section, we now prove a theorem that will give us a threshold function for all graph properties of the form $\mathcal{P}_H$, the property of containing a copy of a fixed graph H as a subgraph.

Let H be given, put $k := |H|$ and $\ell := \|H\|$, and assume that $\ell \geq 1$. Write $X(G)$ for the number of subgraphs of a graph G that are isomorphic to H .

Given $n \in \mathbb{N}$, let $\mathcal{H}$ denote the set of all copies of H on subsets of $\{0, \dots, n-1\}$, the vertex set of the graphs in $\mathcal{G}(n, p)$:

$$\mathcal{H} := \{H' \mid H' \cong H, V(H') \subseteq \{0, \dots, n-1\}\}.$$

Given $H' \in \mathcal{H}$ and $G \in \mathcal{G}(n, p)$, we shall write $H' \subseteq G$ to express that H' itself – not just an isomorphic copy of H' – is a subgraph of G . As in the proof of Lemma 11.1.4, double counting gives

$$\mathbb{E}(X) = \sum_{H' \in \mathcal{H}} \mathbb{P}[H' \subseteq G]. \quad (1)$$

And for every fixed $H' \in \mathcal{H}$ we have

$$\mathbb{P}[H' \subseteq G] = p^\ell, \quad (2)$$

because $\|H\| = \ell$.

Let h denote the number of isomorphic copies of H on a fixed k -set; clearly, $h \leq k!$. As there are $\binom{n}{k}$ possible vertex sets for the graphs in $\mathcal{H}$, we thus have

$$|\mathcal{H}| = \binom{n}{k} h \leq \binom{n}{k} k! \leq n^k. \quad (3)$$

Given a probability $p = p(n)$ and a candidate $t = t(n)$ for a threshold function, we write $\gamma := p/t$. Our first lemma treats the case of $\gamma \rightarrow 0$:

Lemma 11.4.3. *If $t = n^{-1/\varepsilon(H)}$, and p is such that $\gamma \rightarrow 0$ as $n \rightarrow \infty$, then almost no $G \in \mathcal{G}(n, p)$ lies in $\mathcal{P}_H$.*

Proof. Our aim is to find an upper bound for $\mathbb{E}(X)$, and to show that this tends to zero as $n \rightarrow \infty$. By our choice of t and definition of γ we have

$$p = \gamma t = \gamma n^{-k/\ell},$$

so

$$\mathbb{E}(X) \stackrel{(1,2)}{=} |\mathcal{H}| p^\ell \stackrel{(3)}{\leq} n^k (\gamma n^{-k/\ell})^\ell = \gamma^\ell.$$

Thus if $\gamma \rightarrow 0$ as $n \rightarrow \infty$, then

$$\mathbb{P}[G \in \mathcal{P}_H] = \mathbb{P}[X \geq 1] \leq \mathbb{E}(X) \leq \gamma^\ell \xrightarrow{n \rightarrow \infty} 0$$

by Markov's inequality (11.1.5). $\square$

Unlike the function t in Lemma 11.4.3, our threshold function for $\mathcal{P}_H$ will not be expressed in terms of $\varepsilon(H)$, but of

$$\varepsilon'(H) := \max \{ \varepsilon(H') \mid H' \subseteq H \}. \quad \varepsilon'(H)$$

Our second lemma treats the case of $\gamma \rightarrow \infty$:

Lemma 11.4.4. *If $t = n^{-1/\varepsilon'(H)}$, and p is such that $\gamma \rightarrow \infty$ as $n \rightarrow \infty$, then almost every $G \in \mathcal{G}(n, p)$ lies in $\mathcal{P}_H$.*

Proof. By our new choice of t and definition of γ we now have

$$p = \gamma n^{-1/\varepsilon'}, \quad (4)$$

where $\varepsilon' := \varepsilon'(H)$. ε'

Before we start on the main proof, let us note an inequality for later use. For all $n \geq k$,

$$\begin{aligned} \binom{n}{k} n^{-k} &= \frac{1}{k!} \left(\frac{n}{n} \cdots \frac{n-k+1}{n} \right) \\ &\geq \frac{1}{k!} \left(\frac{n-k+1}{n} \right)^k \\ &= \frac{1}{k!} \left(1 - \frac{k-1}{n} \right)^k. \end{aligned} \quad (5)$$

When n gets large and k is bounded, as in our case, the upshot is that n^k exceeds $\binom{n}{k}$ by no more than a constant factor, one independent of n .

Our goal is to apply Lemma 11.4.2, and hence to bound $\sigma^2/\mu^2 = (\mathbb{E}(X^2) - \mu^2)/\mu^2$ from above. To help us estimate $\mathbb{E}(X^2)$, we begin by rewriting X^2 in a strange way, as

$$X^2(G) = |\{H \in \mathcal{H} : H \subseteq G\}|^2 = |\{(H', H'') \in \mathcal{H}^2 : H' \subseteq G \text{ and } H'' \subseteq G\}|.$$

We can now calculate $\mathbb{E}(X^2)$ by double counting, just as in (1):

$$\mathbb{E}(X^2) = \sum_{(H', H'') \in \mathcal{H}^2} \mathbb{P}[H' \cup H'' \subseteq G]. \quad (6)$$

Let us then calculate these probabilities $\mathbb{P}[H' \cup H'' \subseteq G]$. Given $H', H'' \in \mathcal{H}$, we have

$$\mathbb{P}[H' \cup H'' \subseteq G] = p^{2\ell - \|H' \cap H''\|}.$$

As $\|H' \cap H''\| \leq i\varepsilon'$ for $i := |H' \cap H''|$ by definition of ε' , this yields

$$\mathbb{P}[H' \cup H'' \subseteq G] \leq p^{2\ell - i\varepsilon'}. \quad (7)$$

We have now estimated the individual summands in (6); what does this imply for the sum as a whole? Since (7) depends on the parameter $i = |H' \cap H''|$, we partition the range $\mathcal{H}^2$ of the sum in (6) into the subsets

$$\mathcal{H}_i^2 := \{(H', H'') \in \mathcal{H}^2 : |H' \cap H''| = i\}, \quad i = 0, \dots, k,$$

and calculate for each $\mathcal{H}_i^2$ the corresponding sum

$$A_i := \sum_i \mathbb{P}[H' \cup H'' \subseteq G]$$

by itself. (Here, as below, we use $\sum_i$ to denote sums over all pairs $(H', H'') \in \mathcal{H}_i^2$.)

If $i = 0$ then H' and H'' are disjoint, so the events $H' \subseteq G$ and $H'' \subseteq G$ are independent. Hence,

$$\begin{aligned} A_0 &= \sum_0 \mathbb{P}[H' \cup H'' \subseteq G] \\ &= \sum_0 \mathbb{P}[H' \subseteq G] \cdot \mathbb{P}[H'' \subseteq G] \\ &\leq \sum_{(H', H'') \in \mathcal{H}^2} \mathbb{P}[H' \subseteq G] \cdot \mathbb{P}[H'' \subseteq G] \\ &= \left(\sum_{H' \in \mathcal{H}} \mathbb{P}[H' \subseteq G] \right) \cdot \left(\sum_{H'' \in \mathcal{H}} \mathbb{P}[H'' \subseteq G] \right) \\ &\stackrel{(1)}{=} \mu^2. \end{aligned} \quad (8)$$

Let us now estimate A_i for $i \geq 1$. Note that $\sum_i$ can be written as $\sum_{H' \in \mathcal{H}} \sum_{H'' \in \mathcal{H}: |H' \cap H''| = i}$. For fixed H' , the second sum ranges over

$$\binom{k}{i} \binom{n-k}{k-i} h$$

summands: the number of graphs $H'' \in \mathcal{H}$ with $|H'' \cap H'| = i$. Hence, for all $i \geq 1$ and suitable constants c_1, c_2 independent of n ,

$$\begin{aligned}
 A_i &= \sum_i \mathbb{P}[H' \cup H'' \subseteq G] \\
 &\stackrel{(7)}{\leq} \sum_{H' \in \mathcal{H}} \binom{k}{i} \binom{n-k}{k-i} h p^{2\ell} p^{-i\varepsilon'} \\
 &\stackrel{(4)}{=} |\mathcal{H}| \binom{k}{i} \binom{n-k}{k-i} h p^{2\ell} (\gamma n^{-1/\varepsilon'})^{-i\varepsilon'} \\
 &\leq |\mathcal{H}| p^\ell c_1 n^{k-i} h p^\ell \gamma^{-i\varepsilon'} n^i \\
 &\stackrel{(1,2)}{=} \mu c_1 n^k h p^\ell \gamma^{-i\varepsilon'} \\
 &\stackrel{(5)}{\leq} \mu c_2 \binom{n}{k} h p^\ell \gamma^{-i\varepsilon'} \\
 &\stackrel{(1,2,3)}{=} \mu^2 c_2 \gamma^{-i\varepsilon'} \\
 &\leq \mu^2 c_2 \gamma^{-\varepsilon'}
 \end{aligned}$$

if $\gamma \geq 1$. By definition of the A_i , this implies with $c_3 := k c_2$ that

$$\mathbb{E}(X^2)/\mu^2 \stackrel{(6)}{=} \left(A_0/\mu^2 + \sum_{i=1}^k A_i/\mu^2 \right) \stackrel{(8)}{\leq} 1 + c_3 \gamma^{-\varepsilon'}$$

and hence

$$\frac{\sigma^2}{\mu^2} = \frac{\mathbb{E}(X^2) - \mu^2}{\mu^2} \leq c_3 \gamma^{-\varepsilon'} \xrightarrow{\gamma \rightarrow \infty} 0,$$

since $\varepsilon' \geq \varepsilon > 0$ by our assumption that $\ell = \|H\| > 0$. By Lemma 11.4.2, therefore, $X > 0$ almost surely, i.e. almost every $G \in \mathcal{G}(n, p)$ has a subgraph isomorphic to H and hence lies in $\mathcal{P}_H$. $\square$

Theorem 11.4.5. (Erdős & Rényi 1960; Bollobás 1981)

Let H be a graph with at least one edge. Then $t = n^{-1/\varepsilon'(H)}$ is a threshold function for $\mathcal{P}_H$.

Proof. We have to show that almost no $G \in \mathcal{G}(n, p)$ lies in $\mathcal{P}_H$ if $\gamma \rightarrow 0$ as $n \rightarrow \infty$, and that almost all $G \in \mathcal{G}(n, p)$ lie in $\mathcal{P}_H$ if $\gamma \rightarrow \infty$ as $n \rightarrow \infty$. This latter assertion was proved in Lemma 11.4.4.

To prove that almost no $G \in \mathcal{G}(n, p)$ lies in $\mathcal{P}_H$ if $\gamma \rightarrow 0$, we apply Lemma 11.4.3 to a subgraph $H' \subseteq H$ for which the maximum in the definition of $\varepsilon'(H)$ is attained, i.e. which is such that $\varepsilon(H') = \varepsilon'(H)$. The lemma implies that almost no $G \in \mathcal{G}(n, p)$ contains a copy of H' . Since any graph containing H also contains H' , this implies that almost no $G \in \mathcal{G}(n, p)$ contains a copy of H , as desired. $\square$

The bound in Theorem 11.4.5 is particularly easy to compute for *balanced* graphs H , those for which $\varepsilon'(H) = \varepsilon(H)$. Cycles and trees are examples of balanced graphs. For cycles, we have the threshold familiar from the proof of Erdős's theorem:

Corollary 11.4.6. *If $k \geq 3$, then $t(n) = n^{-1}$ is a threshold function for the property of containing a k -cycle.* $\square$

Note that t does not depend on k . (See also Exercise 18.)

For trees, there is a similar phenomenon. Here, the threshold function does depend on the order of the tree, but not on its shape:

Corollary 11.4.7. *If T is a tree of order $k \geq 2$, then $t(n) = n^{-k/(k-1)}$ is a threshold function for the property of containing a copy of T .* $\square$

The systematic study of threshold functions has led to an overall picture of how the typical properties of a graph $G \in \mathcal{G}(n, p)$ unfold as the growth rate of $p = p(n)$ increases. This picture, dubbed the *evolution of random graphs*, is quite fascinating: as in the evolution of species, changes happen 'in jumps', marked by the times the growth rate of p crosses that of a threshold function.

For a very rough sketch, let us begin with edge probabilities p whose order of magnitude lies below n^{-2} ; for such p , a random graph $G \in \mathcal{G}(n, p)$ almost surely has no edges at all. But as p grows, it acquires more and more structure. As p approaches n^{-1} , its components become larger and larger trees (Corollary 11.4.7), until at $p = n^{-1}$ the first cycles are born (Exercise 18). Soon, some of these will have several crossing chords, making the graph non-planar. At the same time, one component outgrows the others, until it devours them around $p = (\log n)n^{-1}$, making the graph connected. Hardly later, at a mere $p = (1 + \epsilon)(\log n)n^{-1}$, our random graph already almost surely has a Hamilton cycle...

Exercises

1. What is the probability that a random graph in $\mathcal{G}(n, p)$ has exactly m edges, for $0 \leq m \leq \binom{n}{2}$ fixed?
2. What is the expected number of edges in $G \in \mathcal{G}(n, p)$?
3. What is the expected number of K^r -subgraphs in $G \in \mathcal{G}(n, p)$?
4. Characterize the graphs that occur as a subgraph in every graph of sufficiently large average degree.